MILITARY SPECIFICATION

CARTRIDGE, 4.2 INCH, HE, M329A2 W/SUPPLEMENTARY CHARGE LOADING, ASSEMBLING AND PACKING

This specification is approved for use by the U.S. Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification contains requirements in addition to those specified on the drawings, quality assurance provisions and packing for one type of high explosive mortar ammunition designated as Cartridge, 4.2 Inch, HE, M329A2.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2) the following specifications and standards of the issue listed in the issue of the Departmnet of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

FSC 1315

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-410	- Non-destructive Testing Personnel Qualification and Certification (Eddy Current, Liquid Penetrant, Magnetic Particle, Radiographic and Ultrasonic)
MIL-STD-453	- Inspection, Radiographic

2.1.2 Other Government documents, drawings and publications. The following Government documents, drawings and publications form a part of this specification to the extent specified herein.

DRAWINGS (See 6.4)

US ARMY RESEARCH AND DEVELOPMENT COMMAND (ARRADCOM))

PRODUCT AND PACKAGING DRAWINGS

9235654	- Cartridge, 4.2-Inch, HE, M329A2, W/Supplementary Charge
9243973	- Packing and Marking for Box, Wood, for 4.2-Inch Mortar Ammunition in Fiber Container, PA46; in Jungle Wrap
9272389	- Packing and Marking for Container, Fiber, PA46 Mod: For 4.2-Inch Mortar Ammunition and Jungle Wrap

INSPECTION EQUIPMENT DRAWINGS

9278618	- Alignment Gage
9278619	- Alignment Check Gage

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer).

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.1.1 Clean explosive scrap. Clean scrap (see 6.7) produced in regular operations may be re-used, provided that it complies with the requirements for the grade of high explosive specified, except for form. Scrap from floor sweeping, ventilation and suction apparatus (other than from drilling operations specifically controlled to prevent contamination of explosive) shall not be used.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on drawing (dwg.) 9235654 and associated drawings and with all requirements specified in applicable specifications and standards.

3.3 Cavitation in high explosive charge.

3.3.1 Projected areas and dimensions. Cavities and cracks within the explosive charge shall not exceed the requirements specified in Table 1 when determined as specified in 4.5.1. Cavities having a maximum (max.) projected length of 1/32 inch or less shall be disregarded, provided they are separated by 1/2 inch of solid cast.

TABLE I. Cavitation requirements.

	SEGMENTS (See Figure 1)			
	A	B	C	D
Sum of projected area of the cavities, excluding pipes, cracks and annular rings square inch (sq. in.)	*1/64	1/4	1/2	1/2
Projected length of any cavity excluding pipes, cracks and annular rings (in.)	1/8	1/2	1/2	3/4
Piping cavities, max projected area sq. in.)	0	1/4	1/2	NA
Piping cavities, max. projected width (in.)	0	1/2	1/2	NA

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Cracks, max. projected width (in.)	1/32	1/32	1/16	NA
Annular rings, max. projected width (in.)	0	0	1/4	NA

* If the length of the largest cavity is 1/16 in. or less, the max. total projected area may be 1/20 sq. in.

NA - not applicable

3.3.2 Porous areas. Porous areas shall be treated as cavities and shall be subject to the restrictions placed upon cavities, except that 80 percent of the projected length and 80 percent of the projected area shall be considered.

3.3.3 Cracked charges. Not more than one transverse crack shall be permitted in Segment A and not more than a total of four transverse cracks shall be permitted in Segment A, B and C, combined. Cracks shall be disregarded in Segment D.

3.3.4 Annular rings. The presence of annular rings 1/16 inch max. and extending from the shell wall 1/4 inch max. into the charge shall be disregarded. The cavity area represented by such annular rings shall not be included when determining cavity area in the charge.

3.3.5 Cavities extending into two segments. The entire area of cavities extending into two segments shall meet the requirements of both segments.

3.3.6 Base separation. The separation between the base of the projectile and the explosive charge shall not be greater than .015 in.

3.4 Surface cavitation. Cavities formed in Segment D due to chipped or broken explosive from the sidewall of the fuze well shall not extend in aggregate around more than 1/4 of the circumference. Cavities in the base of the fuze well shall not be cause for rejection provided the sum of the areas of the cavities is not greater than 80 percent of the area of the fuze well base and no individual point extends more than 1/4 inch below the max. depth of the fuze well permitted by the dwg. and provided no point extends above the flat surface on which the liner would rest. No individual point of explosive shall extend above the flat surface on which the liner would rest. Cavities having a max. dimension of 1/32 inch or less shall be disregarded in the computation. Disregard any cavitation in fuze well above Segment D.

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3.5 Proving ground. The cartridges shall comply with the following requirements when tested as specified in 4.5.2.

3.5.1 Misfire. The cartridge shall not fail to fire from the weapon.

3.5.2 Duds. The cartridge shall function high order on impact.

3.5.3 Short round. The cartridge shall not have a horizontal range less than 80 percent of the average horizontal range (calculated without short rounds.)

3.5.4 Range probable error (see 6.5).

3.5.4.1 Increment charge 3. The horizontal range probable error at increment charge 3 (includes bag loading assembly and three propellant increments) shall not be greater than 3 percent of the mean range.

3.5.4.2 Increment charge 34. The horizontal range probable error at increment charge 34 (includes bag loading assembly and 34 propellant increments) shall not be greater than 1 1/2 percent of the mean range.

3.5.5 Premature burst. There shall be no premature burst in the gun bore or in flight.

3.5.6 Metal parts separation. There shall be no metal parts separation in the gun bore or in flight.

3.6 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specifed in the contract.

3.7 Workmanship. The parts and assemblies shall be free from burrs, chips, dirt, grease and corrosion that may impair proper functioning.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

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4.2 CLassification of inspections. The following types of inspection shall be conducted on this item:

a. First article inspection

b. Quality conformance inspection

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of eighty loaded cartridge assemblies (dwg. 9235654).

4.3.2 Inspections to be performed. See MIL-A-48078 and Table II specified herein.

4.3.3 Rejection. See MIL-A-48078.

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TABLE II - First article inspection.

CLASSIFICATION OF DEFECTS & TESTS MIL-C-48087A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Cartridge Assembly				
	Examination for defects	80	100%	3.2	4.4.2.8
	Proving Ground	80	100%	3.5	4.4.3.1/4.5.2

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4.4. Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots shall contain:

Items of one lot interfix from one manufacturer

a. Projectile metal parts assemblies
b. Composition B
c. Cartridge containers
d. Ignition cartridges
e. Striker nut assemblies
f. Obturators
g. Supplementary charge assemblies
h. Bag loading assemblies
i. Fuzes
j. Propellant increments

4.4.2 Examination. See MIL-A-48078. Unless otherwise specified in the Classification of Defects and Test tables, sampling plans and procedures for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48087A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.1	Projectile Loading Assembly, Prior to Loading				9241269
					NEXT HIGHER ASSEMBLY 9235654
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special					
101	Foreign matter in projectile cavity		100%	3.2	Visual
102	Unsolidified pool of paint in projectile cavity		100%	3.2	Visual
Major	None defined				
Minor					
201	Interior protective coating, inadequate		0.65%	3.2	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48087A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Projectile Loading Assembly after Loading, But Prior to Assembly of Liner	1 OF 1	9241269
			NEXT HIGHER ASSEMBLY 9235654

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Explosive on projectile body		100%	3.2	Visual
Special					
a.	Cavitation, cracks or annular rings in Segment A in excess of that permitted in Table I	*	0.065	3.3.1	4.5.1
b.	Cavitation, cracks or annular rings in Segments B, C or D in excess of twice that permitted in Table I	*	0.065	3.3.1	4.5.1
c.	Cracked charge	*	0.065	3.3.3	4.5.1
d.	Base separation	*	0.065	3.3.6	4.5.1
Major					
101	Depth of well, max.		0.40%	3.3.2	Gage
102	Diameter of well, max.		0.40%	3.3.2	Gage
Minor					
201	Cavitation, cracks, or annular rings in Segments B, C or D in excess but not more than twice of that permitted in Table I	*	0.65%	3.3.1	4.5.1
202	Surface cavitation		0.65%	3.4	Visual
203	Improper coverage of silicone soft film on exterior of liner		0.65%	3.2	Visual
204	Explosive on threads		0.65%	3.2	Visual

NOTES:

* The sampling plan and procedure for this test shall be in accordance with MIL-STD-1235 using CSP-1 plan. The sampling frequency Code letter for both Special and Minor defects shall be G.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48087A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Projectile Loading Assembly after Assembling Liner	1 OF 1	9241269
			NEXT HIGHER ASSEMBLY 9235654

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Liner missing		100%	3.2	Visual
2	Liner not fully expanded and not making contact with fuze thread for full 360°		100%	3.2	Visual
3	Explosive extends beyond top of liner		100%	3.2	Visual
Major					
101	Concentricity of inside diameter of liner with pitch diameter of thread		0.40%	3.2	Gage
102	Depth to bottom of liner, min.		0.40%	3.2	Gage
103	Weight		100%	3.2	Gage
104	Liner loose		100%	3.2	Manual
Minor					
201	Evidence of poor workmanship		0.65%	3.7	Visual

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48087A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 11738362
4.4.2.4	Obturator				NEXT HIGHER ASSEMBLY 9235654
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Length		0.40%	3.2	Gage
102	Large outside diameter		0.40%	3.2	Gage
103	Small inside diameter		0.40%	3.2	Gage
104	Hardness		0.40%	3.2	Gage
105	Presence, location and depth of slits		0.40%	3.2	Visual (see 6.9)
Minor					
201	Evidence of flash		0.65%	3.2	Visual
202	Porosity		0.65%	3.2	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48087A (AR)

PARAGRAPH	TITLE		SHEET		DRAWING NUMBER
4.4.2.5	Cartridge Assembly after Assembling Cartridge Container		1 OF 1		9235654
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Cartridge container moves upon application of assembly torque		0.40%	3.2	Gage
Minor	None defined				

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48087A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.6	Cartridge After Assembling Ignition Cartridge and Stiker Nut Assembly				9235654
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Ignition cartridge missing		100%	3.2	Visual
102	Striker nut assembly moves upon application of assembly torque		0.40%	3.2	Gage
Minor	None defined				

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48087A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.7	Cartridge Assembly, before Assembly of Fuze or Plug	1 OF 1	9345654
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Supplementary charge missing		100%	3.2	Visual
102	Supplementary charge inverted		100%	3.2	Visual
Minor					
201	Supplementary charge not removable by hand		0.65%	3.2	Visual

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PARAGRAPH	TITLE		SHEET	DRAWING NUMBER
4.4.2.8	Cartridge, 4.2 Inch, HE, M329A2 W/Supplementary Charge		1 OF 1	9345654
				NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Short round (Proving Ground)	30/50		3.5.3	4.4.3.1.2/4.5.2
2	Premature burst (Proving Ground)	30/50		3.5.5	4.4.3.1.2/4.5.2
3	Metal parts separation (Proving Ground)	30/50		3.5.6	4.4.3.1.2/4.5.2
4	Subcharges not in proper order (gage length of rear subcharge for determination)		100%	3.2	Gage
5	Bag loading assembly missing or damaged to extent that propellant can escape		100%	3.2	Visual
Major					
101	Misfire (Proving Ground)	30/50		3.5.1	4.4.3.1.2/4.5.2
102	Dud (Proving Ground)	30/50		3.5.2	4.4.3.1.2/4.5.2
103	Range probable error (Proving Ground)	30/50		3.5.4	4.4.3.1.2/4.5.2
104	Assembly fails to freely pass through profile and alignment gage		0.40%	3.2	9278618/9278619
105	Obturator seated improperly		0.40%	3.2	Visual
106	Obturator damaged		0.40%	3.2	Visual
107	Fuze moves upon application of assembly torque (if applicable)		0.40%	3.2	Gage
108	Fuze staked improperly (if applicable)		0.40%	3.2	Visual
109	Fuze not fully seated (if applicable)		0.40%	3.2	Visual
110	Rear propellant holder seated improperly		0.40%	3.2	Visual
111	Parts missing		0.40%	3.2	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48087A (AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER 9235654
4.4.2.8	Cartridge, 4.2 Inch, HE, M329A2 W/Supplementary Charge				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
201	Plug fails to loosen upon application of disassembly torque (if applicable)		0.65%	3.2	Gage
202	Marking misleading or unidentifiable, including incorrect color code		0.65%	3.2	Visual
203	Plug missing (if applicable)		0.65%	3.2	Visual
204	Evidence of poor workmanship		0.65%	3.7	Visual

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9272389
4.4.2.9	Unsealed Fiber Container				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Spacer tube missing		0.40%	5.1	Visual
102	Packing stop missing		0.40%	5.1	Visual
103	Stop missing		0.40%	5.1	Visual
104	Steel disc missing		0.40%	5.1	Visual
Minor					
201	Packaging components (filler, inner filler, cushion) missing		0.65%	5.1	Visual
202	Round cannot be removed by hand		0.65%	5.1	Visual

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MIL-C-48087A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9272389
4.4.2.10	Sealed Container				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Container damaged		0.40%	5.1	Visual
102	Sealing strip incomplete or badly wrinkled		0.40%	5.1	Visual
103	Metal end loose or distorted		0.40%	5.1	Visual/Manual
Minor					
201	Length of sealing strip tab		0.65%	5.1	Gage
202	Marking missing, misleading or unidentifiable		0.65%	5.1	Visual
203	Contents loose		0.65%	5.1	Manual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48087A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.11	Fiber Container with Inner Wrap, Prior to Waxing	1 OF 1	9272389
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Wrapper improperly applied		0.40%	5.1	Visual
102	Wrapper ends improperly folded or taped		0.40%	5.1	Visual
Minor					
201	Length of sealing strip tab		0.65%	5.1	Gage

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48087A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.12	Fiber Container in Jungle Wrap	1 OF 1	9272389
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Wax coating incomplete		0.40%	5.1	Visual
Minor					
201	Marking missing or unidentifiable		0.65%	5.1	Visual
202	Wrapper improperly applied on waxed container		0.65%	5.1	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48087A (AR)

PARAGRAPH	TITLE		SHEET		DRAWING NUMBER
4.4.2.13	Sealed Wood Packing Box		1 OF 1		9243973
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Contents exposed		0.40%	5.2	Visual
102	DOD symbol missing or incorrect		0.40%	5.2	Visual
103	Board broken or split		0.40%	5.2	Visual
104	Hardware or strapping missing, broken or loose		0.40%	5.2	Visual
Minor					
201	Marking misleading or unidentifiable		0.65%	5.2	Visual
202	Hardware improperly engaged		0.65%	5.2	Visual
203	Car seal missing, unsealed or improperly positioned		0.65%	5.2	Visual
204	Contents loose		0.65%	5.2	Manual
205	Handle missing or insecure		0.65%	5.2	Visual/Manual

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4.4.3 Testing.

4.4.3.1 Proving ground. The sample cartridge assemblies shall be fired in two groups each in accordance with 4.5.1. Observations shall be made for compliance with the criteria specified in Table III. In addition, if the range probable error (see 6.5) exceeds the requirement the first article/production lot shall be rejected.

4.4.3.1.1 First article. The first article sample shall consist of eighty cartridge assemblies.

4.4.3.1.2 Production.

4.4.3.1.2.1 Beginning with the first lot produced and continuing until three consecutive lots have been found to be acceptable, eighty sample cartridges shall be randomly selected from each lot.

4.4.3.1.2.2 After three consecutive lots have complied with the criteria of 4.4.3.1.2.1, fifty sample cartridges shall be randomly selected from each lot. In the event that three consecutive lots fail to comply with the requirements the acceptance criteria shall revert to 4.4.3.1.2.1.

Table III

Category	Defect	Acceptance Criteria Sample Size 80		Sample Size 50	
		Accept	Reject	Accept	Reject
Critical	Short round	0	1	0	1
Critical	Premature burst	0	1	0	1
Critical	Metal parts separation	0	1	0	1
Major	Misfire	1	2	1	2
Major	Dud	3	4	2	3

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

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4.5 Methods of inspection.

PRECAUTION - This specification covers sampling and testing of chemical, toxic or explosive materials which are potentially hazardous to personnel. Accordingly, it is emphasized that all applicable safety rules, regulations and procedures must be followed in handling and processing these materials.

4.5.1 Radiographic examination. The sample cartridges shall be subjected to radiographic examination in a base down position. The radiographic beam shall form a 90° angle with the longitidual axis of the projectile at the base of the explosive charge. Examination of the radiographic shall be made to determine compliance with applicable requirements. Radiographic equipment, operations and procedures shall be qualified in accordance with MIL-STD-453. Personnel shall be qualified in accordance with MIL-STD-410. In addition, the sensitivity of the radiographic technique shall be 2.0 percent, and the photographic density of the film used shall be no less than 1.25, as determined in accordance with 4.5.1.2. In the event an original radiograph leaves doubt as to acceptability, the cartridge shall be re-radiographed at an angle 90° from that of the original exposures and in the same plane.

4.5.1.1 Sensitivity check of radiographic technique. The sensitivity of radiographic technique shall be determined by the means of a penetrameter fabricated as specified herein. Three penetrameters shall be taped to each projectile so that the penetrameter holes are located on the axis of the projectile as shown in Figure I. The projectile, with attached penetrameters, shall be positioned between the radiographic source and the radiographic film with the penetrameter facing the radiographic source. When radiographing a group of projectiles on a rotating ring, the three penetrameters, taped to a projectile, may be used to represent the group when checking sensitivity. The sensitivity of radiographic technique shall be considered satisfactory when the image of the penetrameter, including the holes and the outline of the entire internal cavity and the fuze well of the projectile shall be clearly defined on the radiograph.

The penetrameters shall be made of steel having an approximate specific gravity of 7.8. Penetrameter thickness shall be .025 plus or minus .010 inch. Each penetrameter shall have a circular hole with a diameter twice that of the penetrameter thickness. For further details on the construction of the penetrameter see MIL-STD-453. Frequency of sensitivity check shall be no less than one in each fifty radiographs.

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4.5.1.2 Check of photographic density of film. The photographic density of the film shall be determined on the film at the point of lightest film density within the area of the explosive, and a calibrated photoelectric densitometer shall be used. Film density shall be checked each hour of operation, or fraction thereof, for each film processor. Greater frequency shall be required for any period during which difficulty is encountered in establishing the required densities.

4.5.1.3 Radiographic standard for base separation. The radiographic standard for base separation shall be developed by the contractor and approved by DRDAR-QAT-A prior to submission of the first article sample. The standard shall be utilizing the aft end of the projectile, cut-off at the forward end of the rotating band. A removable plug of epoxy shall be cast in the cut-off projectile. Radiographs at simulated base separations of .010, .015 and .020 inch shall be taken in order to bracket the simulated base separation of .015 inch for sensitivity. The radiographic standard shall be verified at a minimum by the daily radiograph of the .015 inch simulated base separation cut-off projectile. Acceptance of loaded projectiles shall be made by comparison of the sample radiograph to the standard.

4.5.2 Proving ground test. The rounds shall be conditioned to, and fired at, a temperature of $70^{o} \pm 5^{0}$ Fahrenheit (F), $20^{0} \pm 3^{o}$ Centigrade (C). Warmer rounds shall be fired to attain the approximate gun tube temperature that will be maintained during the test. The number of rounds required for the temperature condition specified are as follows:

Ambient Temp.	Number of Rounds
Above 72°F	2
32°F to 72°F	3
Below 32°F	4

A steady rate of fire shall be maintained throughout the test. If for any reason there is a break in the firing, warmer rounds shall be fired to attain the original gun tube temperature.

The cartridges shall be fired from a 4.2 inch M30 mortar tube. One half of the cartridges shall be fired at increment charge 3 (bag loading assembly plus 3 propellant increments) at an elevation of 1065 mils, and the other half of the cartridges shall be fired at increment charge 34 (bag loading assembly plus 34 propellant increments) at an elevation of 800 mils. All fuzes shall be set for superquick functioning.

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Twenty control rounds will be fired with each set of forty test rounds and fifteen control rounds will be fired with each set of twenty five test rounds, using the bracketing technique.

The individual velocities, ranges and pressures shall be recorded on the firing record.

Observations shall be made for misfires and duds and shall be recorded. Should a premature burst or metal parts separation occur, a photograph shall be taken of the evidence and attached to the firing record. The velocity and range standard deviations and arithmetic averages shall be calculated and recorded. Temperature of the day, weather and the time of firing and time of flight shall also be recorded.

Instrumentation shall be used capable of observing for metal parts separation, flight velocity, position in space vs time, flight anomalies, impact point (range and dispersion) of the control and test rounds. The radar data of all the samples of a rejected lot should be retained for further analysis by DRDAR-QAT-A. Data from acceptable lots may be discarded.

Smear camera shall be used to photograph all test and control rounds. The camera shall be positioned to pick up each round just outside the second velocity coil. The film should be retained by the proving ground for three months, min. at which time it may be discarded. Prints of the film will be made and forwarded for analysis to, and by request of DRDAR-QAT-A.

4.5.2.1 Test validity. If for any reason the proving ground or the technical agency considers that the test conditions have detrementally affected the test results, the test results shall be declared invalid and a new test shall be performed on additional cartridges, as required.

5. PACKAGING

5.1 Packaging requirements. Packaging shall be as specified on dwg. 9272389.

5.2 Packing and marking. Packing and marking shall be as specified on dwg. 9243973.

6. NOTES

6.1 Intended use. The cartridge covered by this specification is intended for use in the 4.2 Inch Mortar System.

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6.2 Ordering data. See MIL-A-48078.

6.3 Submission of inspection equipment for design approvals. See MIL-A-48078. Submit designs as required to Commander, U.S. Army Armament Research and Development Command, ATTN: DRDAR-QAT-I, Dover, NJ 07801.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

6.5 Range probable error. Range probable error is defined as being the range standard deviation multiplied by the factor .6745.

6.6 Submission of test data. One copy each of proving ground ballistic acceptance test data will be forwarded to the following:

Commander
ARRADCOM
ATTN: DRDAR-QAT-B, LCU-SI
Dover, NJ 07801

6.7 Clean scrap. Clean scrap is defined as scrap formed in kettles, pails, risers and loading machines used in the pouring and cooling operations, and scrap formed during the drilling operations.

6.8 Loading of high explosive. Solid explosive shall not be added to the melted explosive or to the projectile body during pouring or cooling.

6.9 Visual examination of slits. When inspecting for slits, the obturator shall be distorted near slots in such a manner so that the slits are visible. Should the depth of the slit be in doubt as a result of visual examinaton, it may be measured or gaged to determine compliance.

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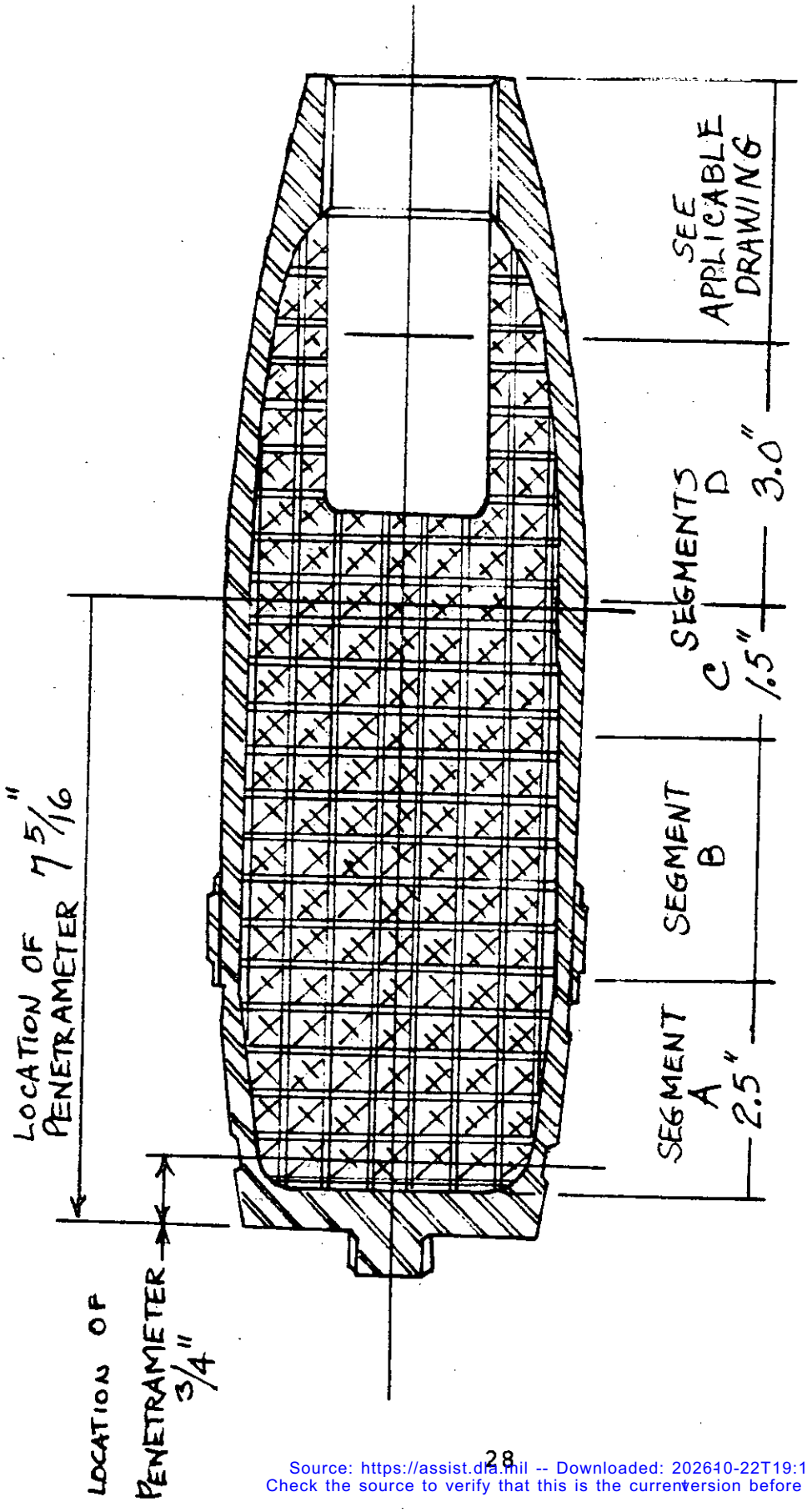


FIGURE 1. Cavitation requirements.

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Custodian:
Army-AR

Preparing activity:
Army-AR

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MIL-C-48087A Cartridge, 4.2 INCH, HE, M329A2 W/Supplementary Charge LAP

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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